

US Used Cars Dataset Executive Summary

Comp 333 – Final Project

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Project Overview

This project contains an end-to-end pipeline and analysis of the Used Cars Dataset (Kaggle). The Used Cars dataset is a real-world collection of approximately 3 million used vehicle listings across the US. The dataset contains nearly 10GB of data, but for the purposes of this analysis a sample of roughly 1GB worth of data was used. This data contained variables such as mileage, year and price. It also required extensive cleaning and feature engineering to fix inconsistent formats and high rates of missing values.

Research Questions

Two research questions were proposed to guide the entire project. One supervised and one unsupervised question:

- **Supervised research question:** Can we predict the selling price of a used vehicle given its specifications, usage history, and dealership characteristics?
- **Unsupervised research question:** Can we discover natural clusters of used vehicles (e.g., budget, mid-range, luxury) based on their specifications and market characteristics?

Approach

The project was split into three progressive phases:

- **Phase 1 – Data Acquisition & Baseline:** Data retrieval, wrangling/cleaning (impute missing values with indicator flags, outlier removal, string normalization, high missingness column removal), Exploratory Data Analysis (EDA) and a baseline linear regression model.
- **Phase 2 – Advanced Modeling:** Advanced supervised learning using models such as Random Forest and XGBoost, feature engineering and feature selection using filter wrapper and embedded methods, and Unsupervised learning with K-Means clustering and PCA dimensionality.
- **Phase 3 – Complete Pipeline and Evaluation:** full pipeline integration into modular functions, Ensemble stacking and anomaly detection, Ethical considerations and a comprehensive analysis across all models.

Key Results

Supervised Learning

The table below summarizes test-set performance across all models. All these tests are based on the feature engineered dataset “feature_cars”. The stacking ensemble achieved the best results on every metric.

Model	R ²	MAE (\$)	RMSE (\$)	Notes
Linear Regression (baseline)	0.7577	3794	4886	Starting reference
Random Forest	0.9219	2029	2775	Strong performance
XGBoost	0.9230	2054	2755	Close to RF
Stacking Ensemble	0.9264	1995	2693	Final selected model

The stacking model’s improvement over Random Forest and XGBoost is modest but consistent across all metrics. These models had the most accurate predictions for mid-range vehicles but struggled more with predicting luxury and budget vehicles.

Unsupervised Learning

K-Means clustering identified distinct vehicle segments based on specifications and market attributes, with price deliberately excluded as a clustering input. Comparing price distributions across clusters revealed meaningfully different median prices. The silhouette score of approximately 0.27 indicates moderate cluster separation. This was the highest silhouette score the algorithm produced and with only two clusters. The silhouette score indicates that there were many overlapping segments and that categories aren’t perfectly distinct.

Conclusion

This project successfully implements a complete, reproducible data analytics pipeline on a large real-world dataset. The supervised models address research question 1 by predicting vehicle price with strong accuracy, while the unsupervised analysis addresses research question 2 by identifying interpretable market segments in feature space. Together, these results provide a consensus of the used-car listings in the US.